HOMEWORK 1 / COMPLETED EXPERIMENTS

Contextual and LLM Bandits

BIOSTAT825 - Experimental Results and Discussion

Student NetID: **zq63** | Submission report: **November 3, 2026**

This report presents the eight completed assignment sections in the order Parts 1-6, 7A, and 7B. All results come from the executed notebook and saved full-size arrays. The original horizon is **T = 1000** throughout. No experiment was rerun or reduced to prepare this report.

Assignment section	Monte Carlo runs N	Report pages
Part 1 - Linear contextual policies	50	2
Part 2 - Epsilon-LinTS selection	50 per comparison	3-4
Part 3 - Linear-model LMCTS	5	5
Part 4 - Three reward functions	5 per environment	6-7
Part 5 - Neural quadratic bandits	10	8
Part 6 - Shuttle and LMCTS tuning	10 per comparison	9-10
Part 7A - LLM movie recommendations	3 per role	11-13
Part 7B - LLM patient preferences	3 per profile	14-15

Evaluation conventions

Parts 1-6 measure cumulative **pseudo-regret** from expected rewards; lower is better. Parts 7A-7B measure cumulative sampled reward; higher is better.

$$R(T) = \Sigma_{t=1}^{T} [\max_a \mu_t(a) - \mu_t(a_t)] \qquad C(T) = \Sigma_{t=1}^{T} r_t$$

Tables report final totals as **mean ± sample standard deviation**. Shaded curve bands are pointwise empirical 10th-90th percentiles across runs, not confidence intervals. Policies start each run with fresh state and use matched environment randomness within each comparison. Rankings describe the tested instances and seeds.

Execution: Duke DCC, one H200, existing RL environment; job 60876090 completed in 2:01:18. All 43 code cells and 16 saved figure pairs passed independent result validation. Reproducibility details and model revisions are on page 16.

Part 1. Linear contextual bandits

The synthetic linear environment has d = 7, three available arms per round, and two fixed context sets. Arm vectors and the unknown coefficient have unit norm. Observed rewards are $x^T\theta$ plus Gaussian noise with standard deviation 1. Greedy, epsilon-greedy (0.1), LinUCB, and LinTS use ridge/prior precision 1.

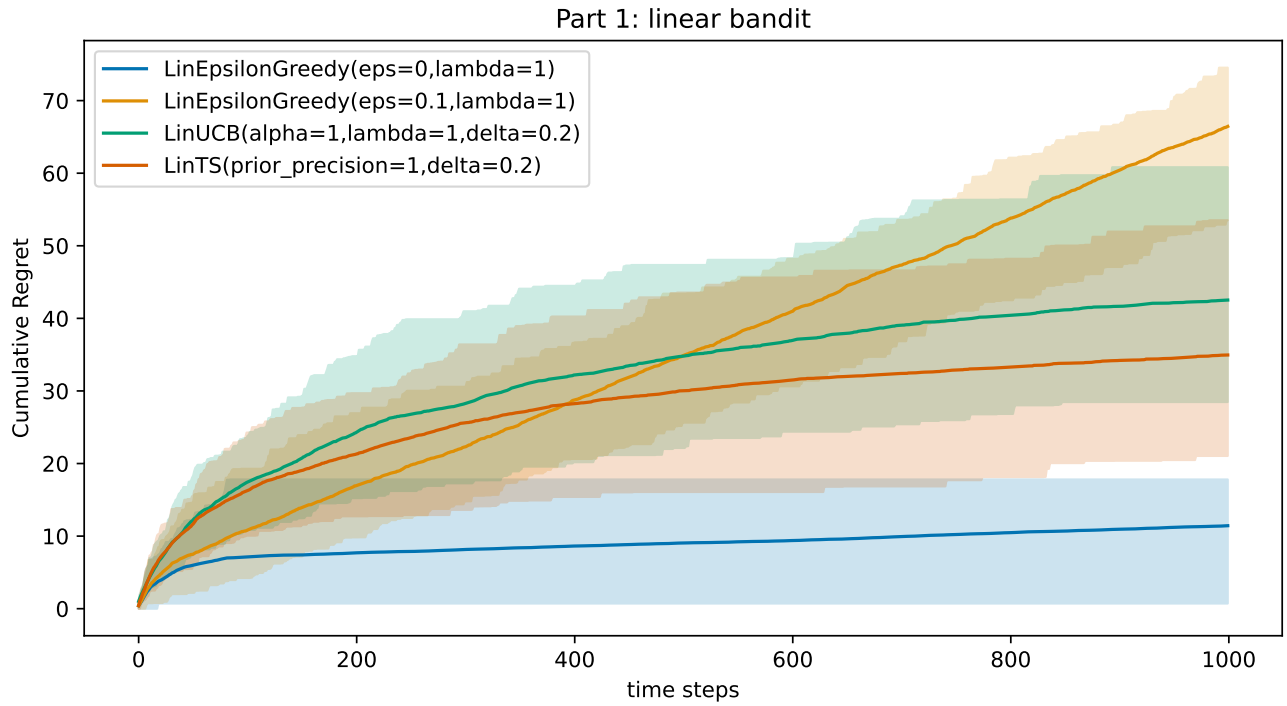


Figure 1. Part 1: linear contextual-bandit policies.

Method	Final cumulative regret
Greedy	11.43 ± 34.59
Epsilon-greedy, epsilon = 0.1	66.45 ± 28.18
LinUCB	42.52 ± 11.60
LinTS	34.95 ± 14.39

Results and discussion

Greedy had the lowest mean regret on this fixed environment, but its high variability matters: the median was 3.23 and the maximum was 237.06. Thus, a low average did not prevent an occasional poor trajectory. LinTS had lower mean regret than LinUCB, whereas LinUCB had the smallest standard deviation. Constant epsilon-greedy exploration had the largest mean regret, consistent with continuing to choose random arms even after learning useful estimates. These results describe trajectories in one generated environment, not a universal ranking across environments. The shaded 10th-90th percentile bands describe run variability, not confidence intervals for the mean.

Part 2. Choosing epsilon for LinTS

Epsilon-LinTS uses a Thompson draw with probability epsilon and the greedy ridge prediction otherwise. The four tested rules are 0.05, 1/d = 1/7, 1/t, and an additional constant 0.2. Time is one-based, so 1/t starts at 1. The sweep reuses the Part 1 environment with N = 50, T = 1000, and seed 62.

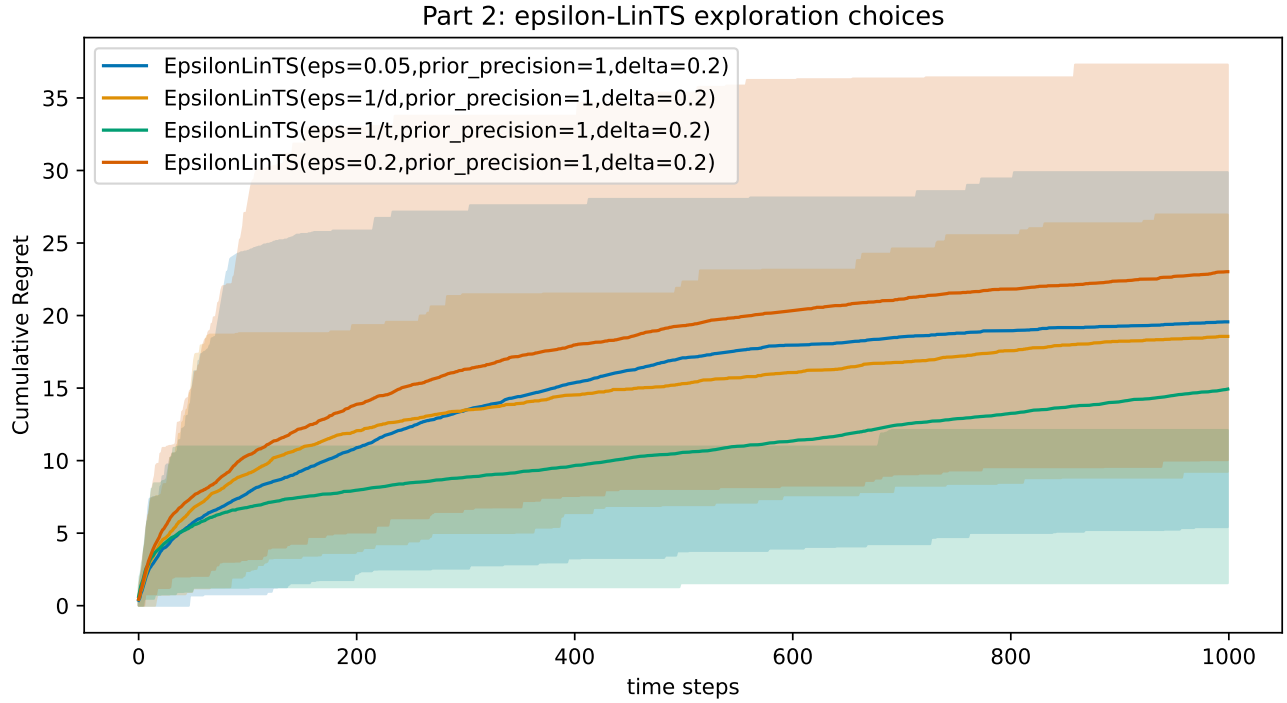


Figure 2. Part 2: epsilon-LinTS exploration-rule sweep.

Epsilon-LinTS rule	Sweep regret (seed 62)
0.05	19.57 ± 32.78
1/d = 1/7	18.56 ± 12.84
1/t	14.93 ± 58.04
0.2 (additional choice)	23.02 ± 21.77

Selection and stability

The code selected **epsilon = 1/t** from the lowest observed sweep mean, then evaluated it against fresh baselines using seed 63.

The selected 1/t rule has the lowest observed sweep mean but a standard deviation of 58.04. By comparison, 1/d has a slightly higher mean (18.56) with a much smaller standard deviation (12.84). Mean-based selection therefore trades average performance against sensitivity to unfavorable trajectories.

Part 2. Fresh baseline comparison

The selected 1/t rule is compared with fresh baseline agents and trajectory seed 63, again using N = 50 and T = 1000. No sweep trajectory is reused as a fresh-comparison result.

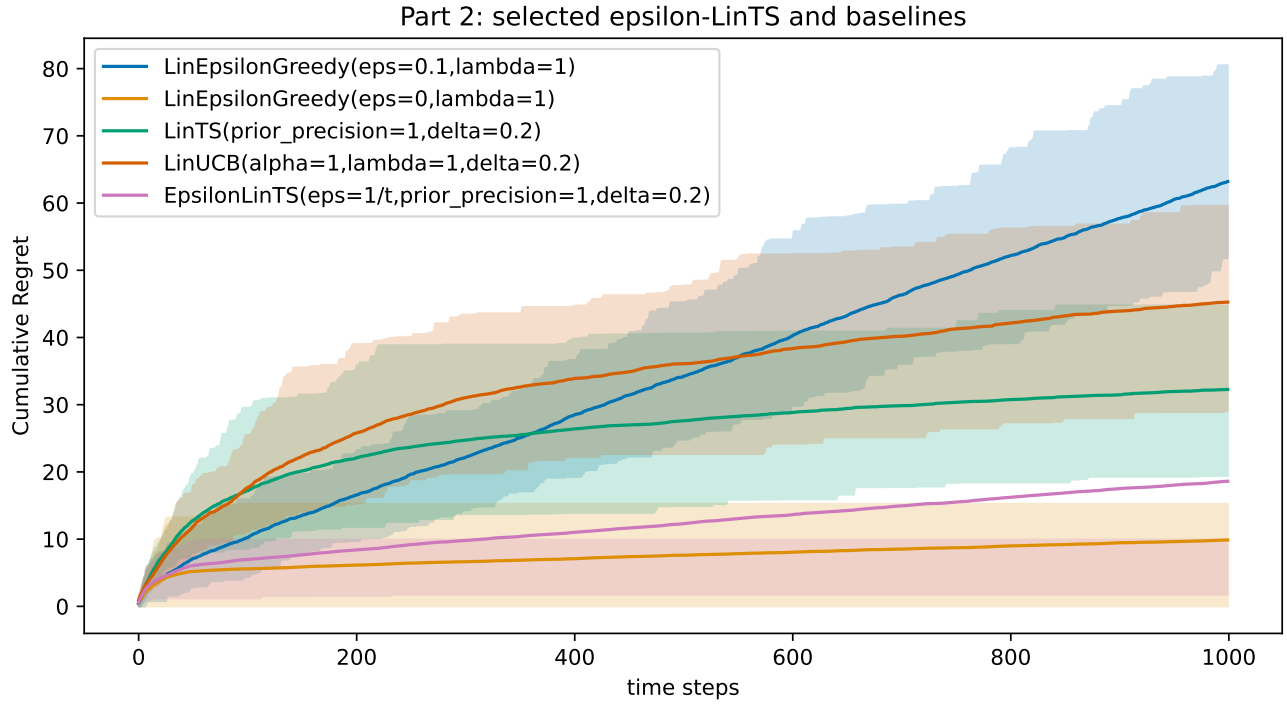


Figure 3. Part 2: selected epsilon-LinTS and fresh baselines.

Method	Final cumulative regret
Greedy	9.87 ± 33.30
Epsilon-greedy, epsilon = 0.1	63.19 ± 11.99
LinTS	32.26 ± 10.34
LinUCB	45.26 ± 13.26
Selected epsilon-LinTS, 1/t	18.61 ± 62.81

Results and discussion

The selected rule had lower mean regret than the three required exploration baselines, while Greedy remained lower. However, 1/t was unreliable in some trajectories: its fresh-comparison median was 4.18, but two runs exceeded 200 and the maximum was 385.11. Rare failures explain its large standard deviation and can place the mean above the 90th percentile. The 1/d rule was more stable in the sweep. Selection identifies the best tested mean on these seeds, not a globally optimal or uniformly reliable epsilon rule.

Part 3. Linear-model LMCTS

A new linear environment uses N = 5 and T = 1000. LMCTS selects the largest current prediction from a bias-free linear model and makes one warm-start Langevin update per observation using all available history. Settings are learning rate 0.1, inverse temperature 0.01, noise multiplier 1, and weight decay 1. The loss is mean MSE.

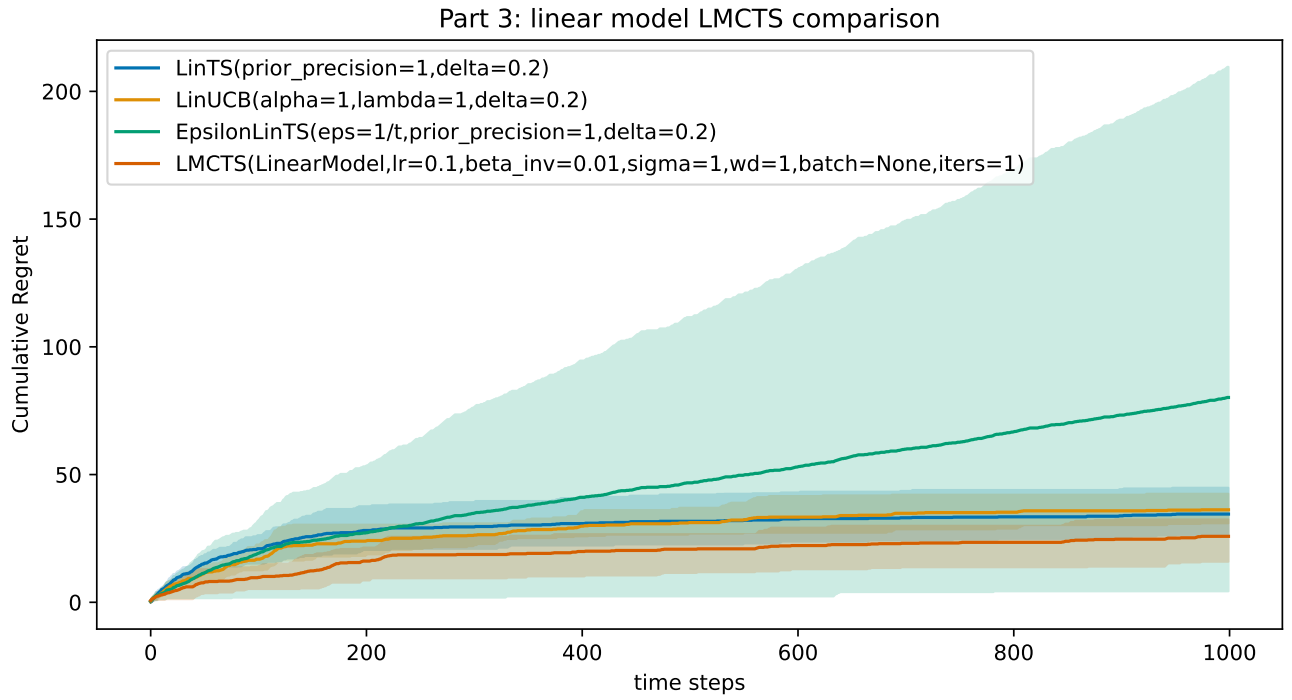


Figure 4. Part 3: linear-model LMCTS and comparison policies.

Method	Final cumulative regret
LinTS	34.54 ± 9.67
LinUCB	36.21 ± 5.79
Epsilon-LinTS, 1/t	80.18 ± 134.96
Linear-model LMCTS	25.82 ± 8.99

Results and discussion

LMCTS had the lowest mean regret in this comparison, followed by LinTS and LinUCB. This is evidence for the supplied LMCTS settings on this instance, rather than a general dominance claim: only five runs were performed. The epsilon rule selected in Part 2 did not transfer reliably to this new environment. Its median regret was 27.88, but one run reached 319.71 and substantially increased the mean. This behavior is consistent with quickly decreasing exploration leaving some early estimation errors difficult to correct. The results should be compared within this experiment because its environment and Monte Carlo seeds differ from Part 2.

These finite-step, regularized-loss Langevin updates are the notebook's explicit implementation assumption; an exact posterior or convergence guarantee is not claimed.

Part 4. Linear and logistic rewards

All three environments share the same six arm-feature vectors and coefficient, with N = 5 and T = 1000. The mean is $x^T\theta$ for linear rewards, sigmoid($x^T\theta$) for Bernoulli logistic rewards, and $10(x^T\theta)^2$ for Gaussian quadratic rewards. Policies and LMCTS settings are the same as in Part 3.

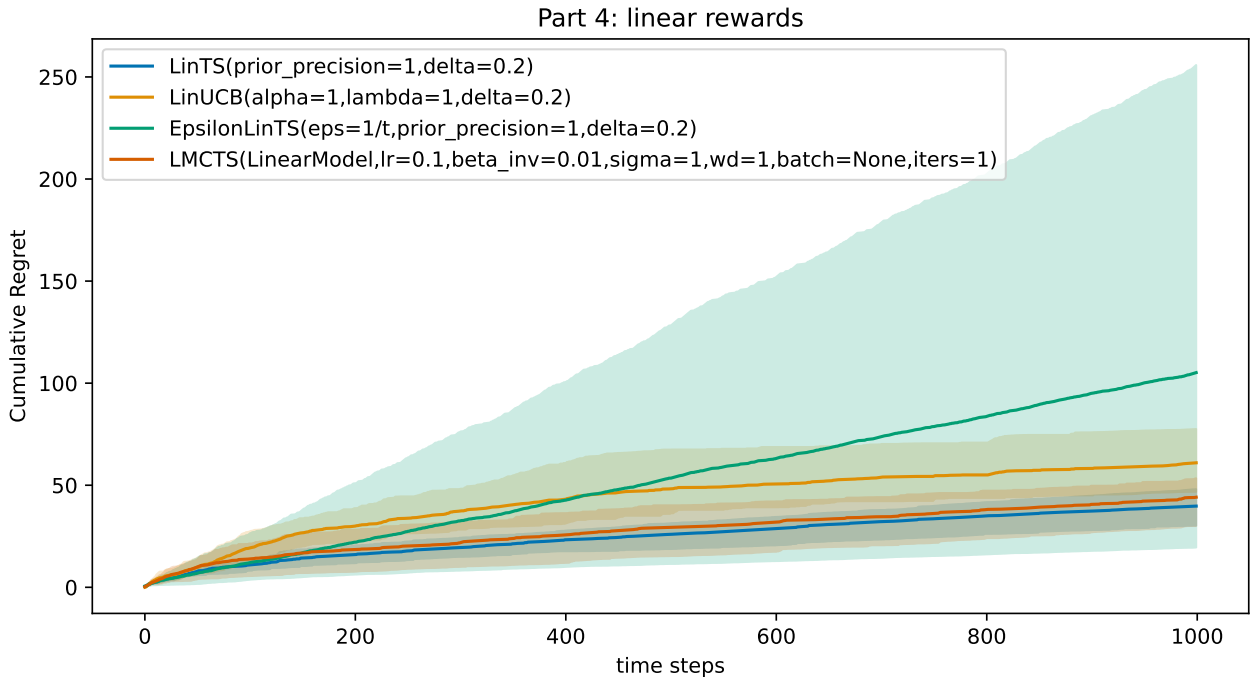


Figure 5. Part 4: shared arm geometry, linear reward function.

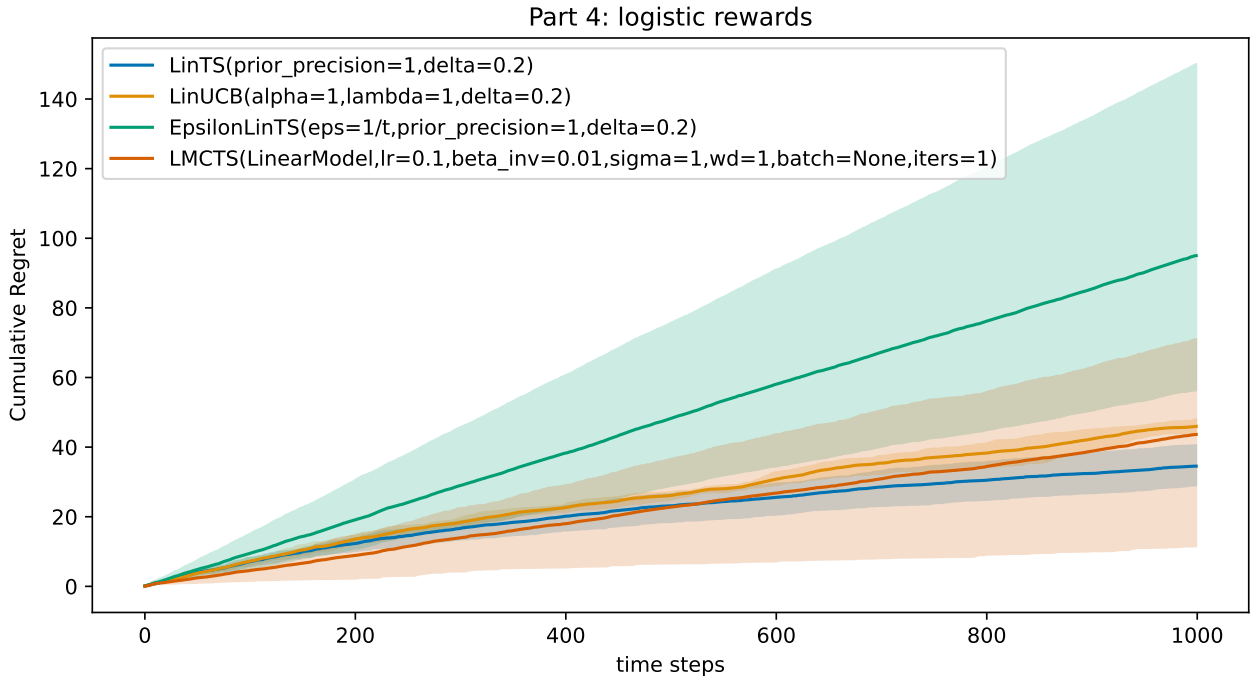


Figure 6. Part 4: shared arm geometry, logistic reward function.

Part 4. Quadratic rewards and comparison

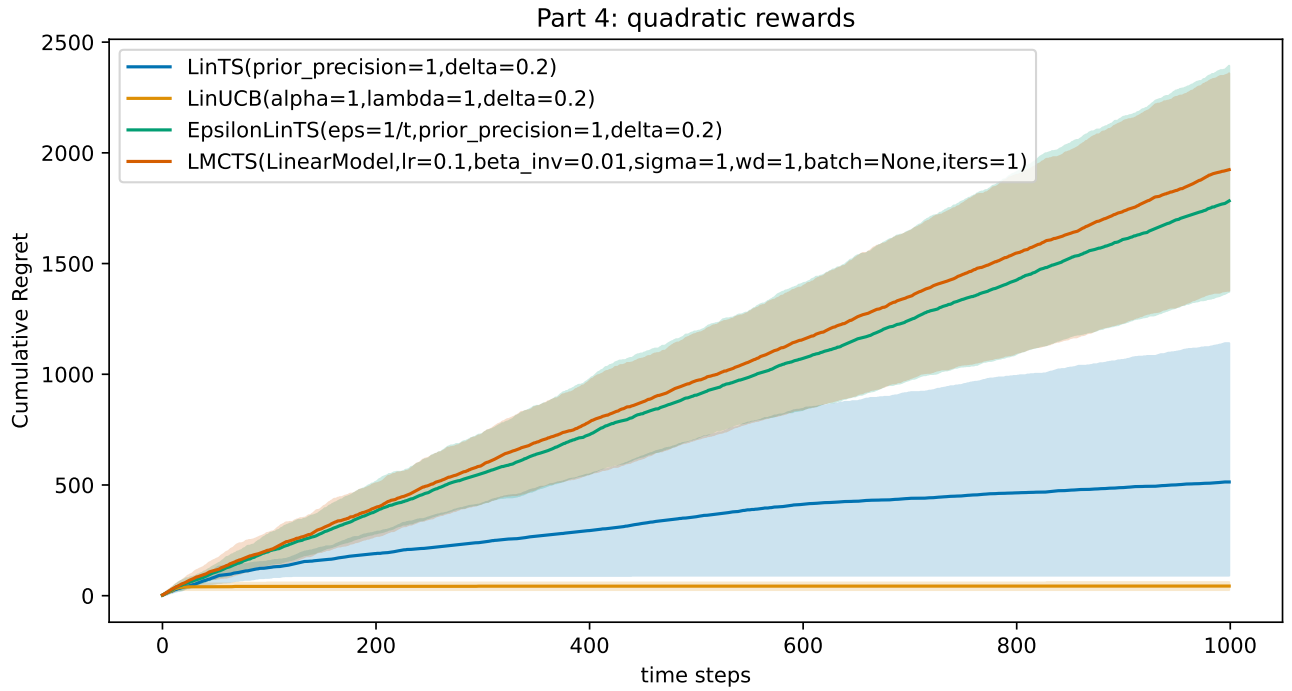


Figure 7. Part 4: shared arm geometry, quadratic reward function.

Method	Linear	Logistic	Quadratic
LinTS	39.78 ± 8.85	34.53 ± 6.13	513.62 ± 556.66
LinUCB	60.99 ± 16.04	45.97 ± 1.98	43.52 ± 17.28
Epsilon-LinTS, 1/t	105.17 ± 164.63	95.00 ± 57.11	1783.07 ± 547.18
Linear-model LMCTS	44.14 ± 13.88	43.63 ± 30.21	1924.01 ± 501.59

Results and discussion

LinTS had the lowest mean regret for linear and logistic rewards. LinUCB was markedly better in the quadratic environment, while LinTS showed high run-to-run variability and epsilon-LinTS and LMCTS accumulated much larger regret. Nonlinear rewards do not automatically imply unavoidable linear approximation error here: only six fixed vectors occur in seven dimensions, allowing a linear model to interpolate their means when those vectors are independent. Regularization, exploration, and finite-step fitting also affect performance; these results do not isolate their separate contributions. Reward scales differ, so compare methods within each column rather than treating raw regret levels across columns as equivalent. With only five trajectories, rankings remain instance-specific.

Part 5. Neural policies for quadratic rewards

Each policy uses an independent scalar-output MLP with one 50-unit LeakyReLU hidden layer. Epsilon-greedy (0.1), NeuralTS, and NeuralUCB use SGD learning rate 0.01; updates use up to 64 historical observations and five steps. NeuralTS/UCB use a diagonal gradient-precision approximation with nu = lambda = 1. NeuralLMCTS uses learning rate 0.01, inverse temperature 0.01, noise multiplier 1, and decay 0.01.

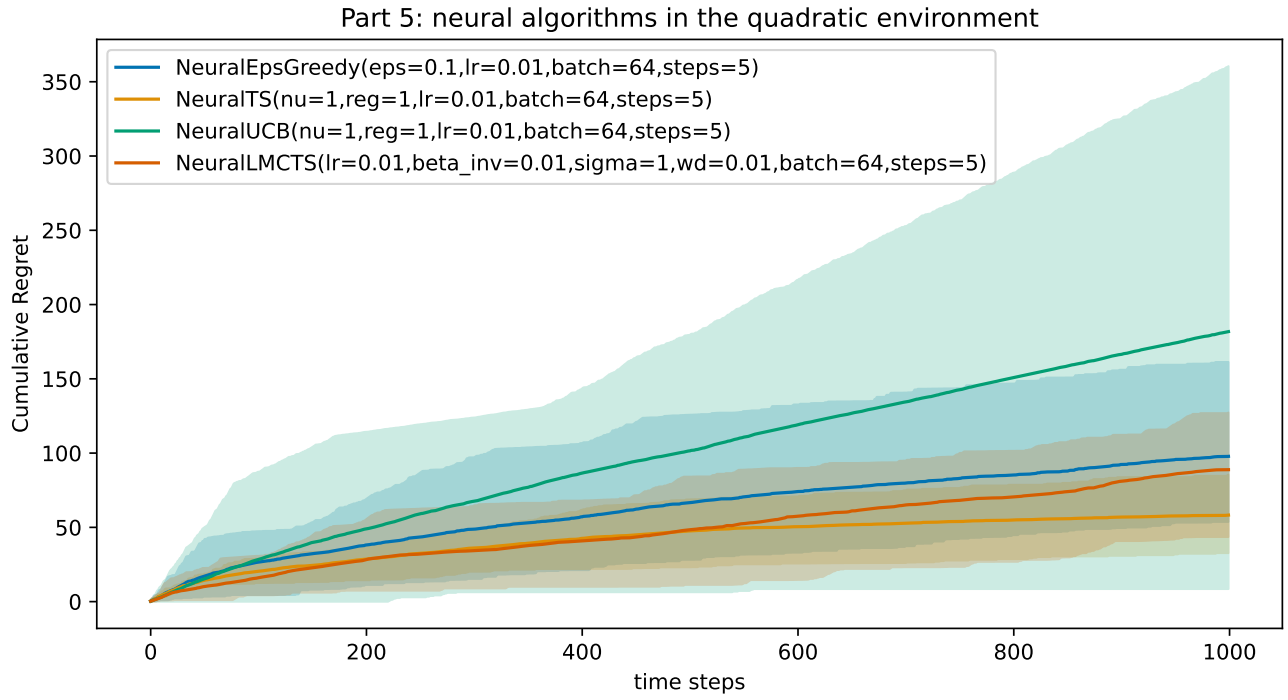


Figure 8. Part 5: neural policies in the quadratic environment.

Method	Final cumulative regret
Neural epsilon-greedy (epsilon = 0.1)	97.81 ± 55.26
NeuralTS	58.26 ± 36.98
NeuralUCB	181.78 ± 310.49
NeuralLMCTS	88.83 ± 43.57

Results and discussion

NeuralTS had the lowest mean regret, followed by NeuralLMCTS and neural epsilon-greedy. NeuralUCB was highly variable: its median was 51.58, but one trajectory reached 1008.40, strongly increasing its mean and standard deviation. Thus, its average should not be read as uniformly poor performance across runs. NeuralLMCTS had lower mean regret than NeuralTS at round 100 (16.00 versus 20.38), but NeuralTS accumulated less additional regret during rounds 501-1000 (10.67 versus 40.46). These results favor NeuralTS on this instance without establishing a universal ranking. Part 5 uses a fresh environment generated with seed 505, so its raw regret cannot establish a controlled neural-versus-linear improvement over Part 4.

Part 6. LMCTS tuning on shuttle

The shuttle classification problem becomes a seven-arm contextual bandit: each nine-dimensional, row-normalized context occupies one of seven disjoint feature blocks (63 inputs). Correct choices earn 1 and other choices earn 0. The four-setting sweep uses seed 606, N = 10, and T = 1000, holding width 50, batch size 64, noise multiplier 1, and weight decay 0.01 fixed.

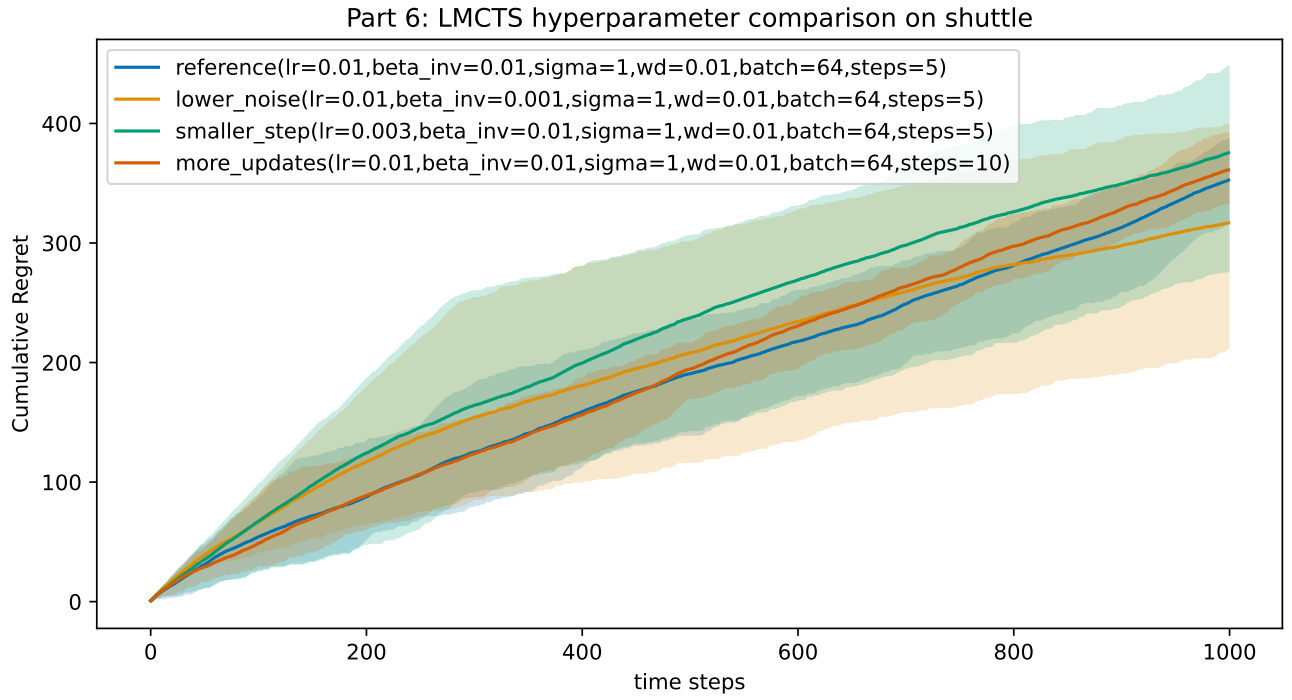


Figure 9. Part 6: four tested LMCTS configurations on shuttle.

LMCTS configuration	Learning rate	Inverse temperature	Updates	Tuning regret
Reference	0.01	0.01	5	352.70 ± 47.75
Lower noise	0.01	0.001	5	316.90 ± 151.41
Smaller step	0.003	0.01	5	375.50 ± 128.20
More updates	0.01	0.01	10	361.10 ± 35.28

Results and discussion

The seed-606 sweep selected **learning rate 0.01, inverse temperature 0.001, and five updates** by its lowest observed mean. Its variability was much larger than the reference configuration's. The ten-update configuration was steadier but did not improve the mean.

Part 6. Fresh neural-policy comparison

The selected configuration (learning rate 0.01, inverse temperature 0.001, five updates) is evaluated against the other neural policies using fresh trajectory seed 607, N = 10, and T = 1000. This uses new online trajectories on the same dataset.

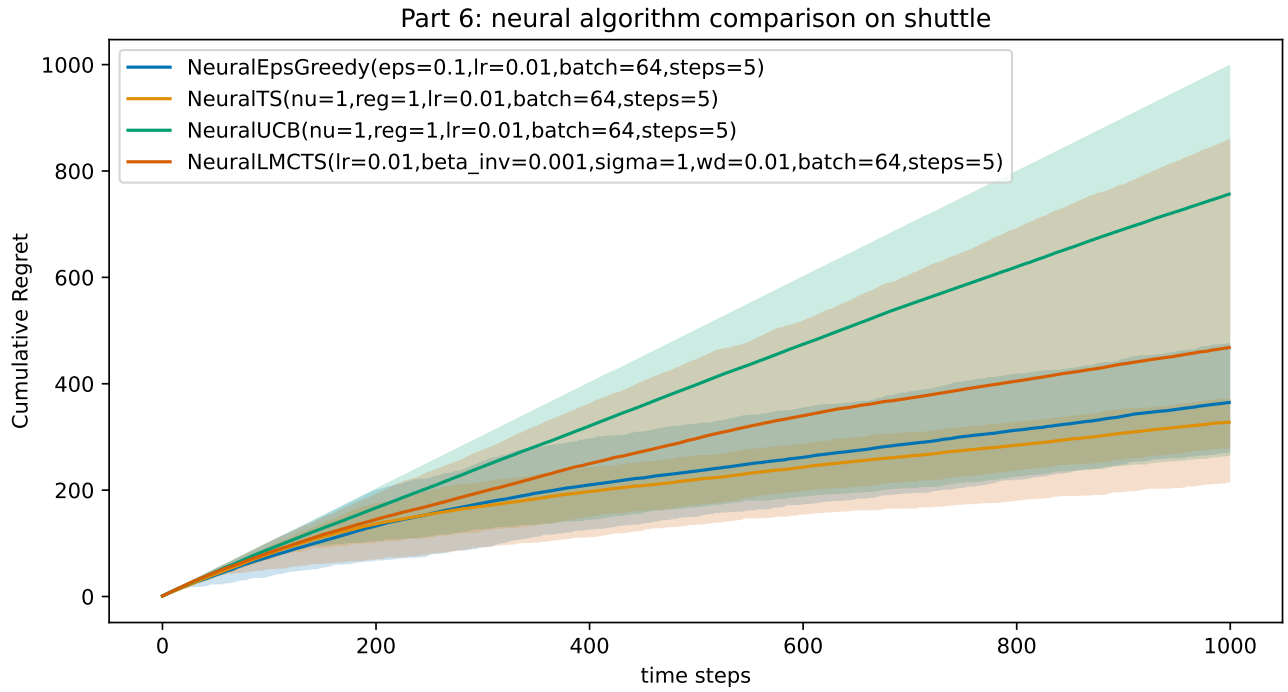


Figure 10. Part 6: fresh comparison with the selected LMCTS setting.

Method, fresh seed 607	Final regret	Mean online accuracy
Neural epsilon-greedy	364.80 ± 84.00	63.52%
NeuralTS	327.70 ± 51.85	67.23%
NeuralUCB	756.50 ± 275.05	24.35%
Selected NeuralLMCTS	468.10 ± 253.10	53.19%

Results and discussion

NeuralTS performed best in the fresh comparison. Selected LMCTS increased from 316.90 tuning regret to 468.10, showing that its favorable selection mean did not transfer reliably to new trajectories. This is not a paired fresh-seed comparison of all LMCTS configurations and does not establish a global optimum. Here regret counts incorrect selections, so online accuracy is 1-R(1000)/1000; it is not held-out supervised test accuracy. The wide LMCTS variation supports reporting repeated-run results rather than recommending the selected setting from its sweep mean alone.

Part 7A. LLM movie recommendations

Qwen3.5-0.8B scores one of eight movies without conversation history. Role A favors space science and engineering; Role B favors romantic musicals. An integer score/10 receives Gaussian noise with standard deviation 0.1 and is clipped to [0, 1]. Sampling temperature is 0.5. Random, Greedy, epsilon-greedy (0.1), and UCB use N = 3 and T = 1000 per role, with seed 7001. Bands are descriptive percentiles.

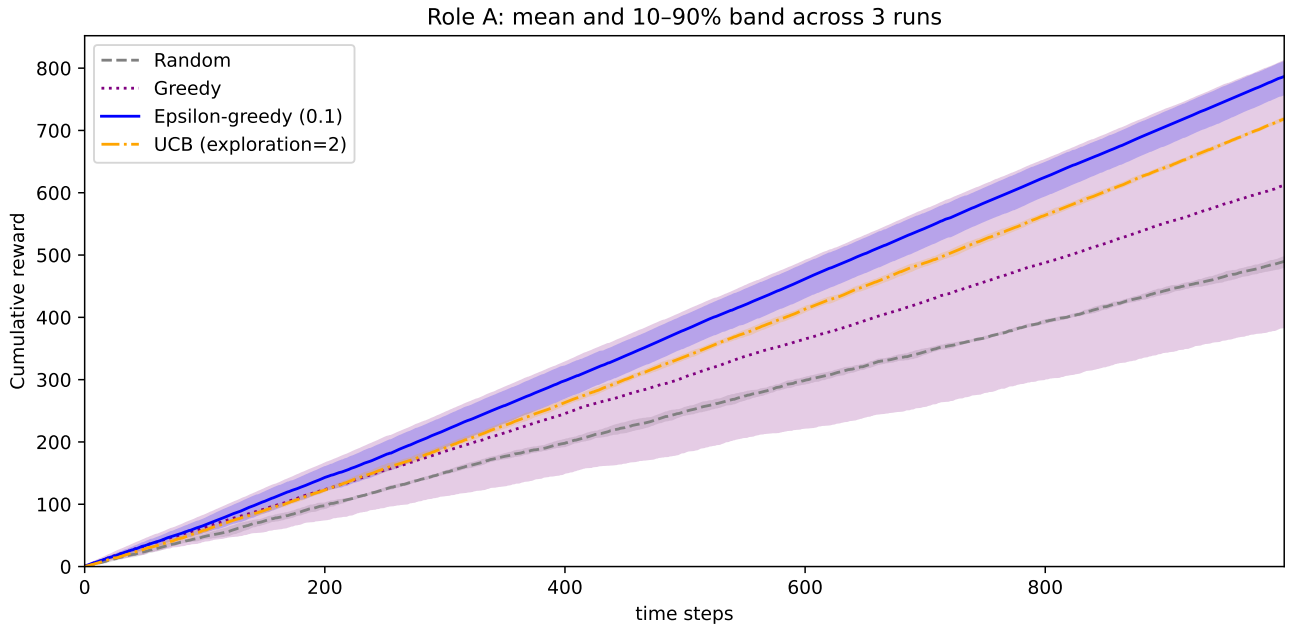


Figure 11. Part 7A: cumulative reward for Role A (space science).

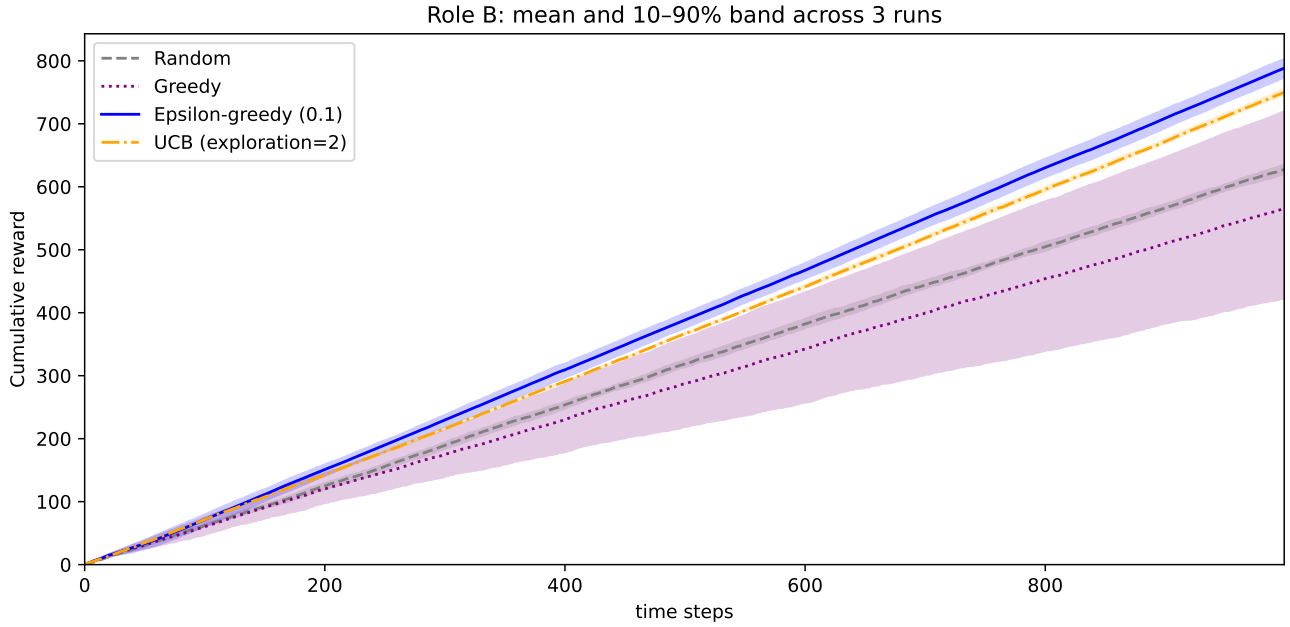


Figure 12. Part 7A: cumulative reward for Role B (romantic musicals).

Part 7A. Rewards and Role A selections

Final cumulative rewards are mean ± sample standard deviation; higher is better.

Policy	Role A	Role B
Random	490.09 ± 10.58	627.69 ± 9.41
Greedy	612.05 ± 273.43	565.43 ± 187.63
Epsilon-greedy (0.1)	786.68 ± 34.34	788.35 ± 18.56
UCB	718.46 ± 2.96	749.91 ± 6.33

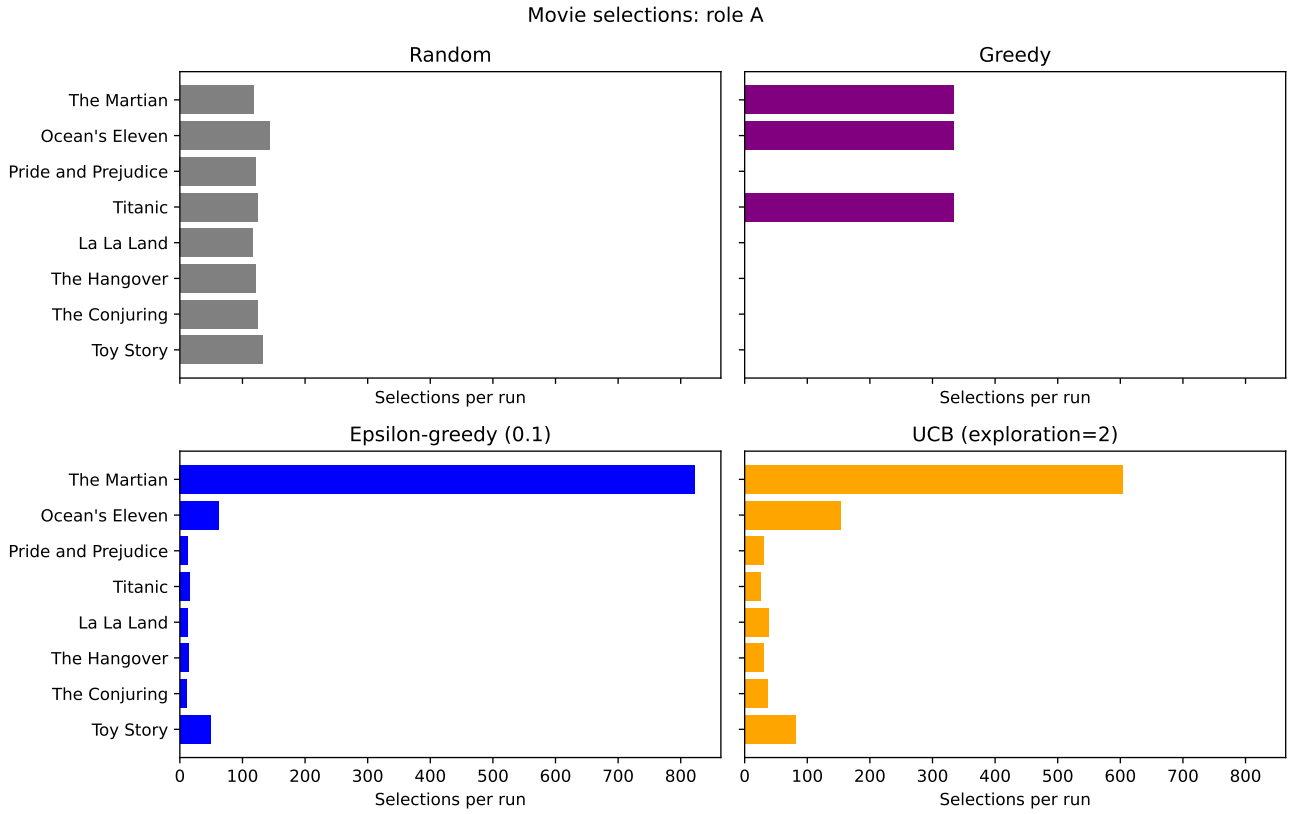


Figure 13. Part 7A: mean movie-selection counts per run for Role A.

Epsilon-greedy achieved the highest mean reward for both roles; UCB had lower means but much smaller between-run variation. For Role A, both favored The Martian, matching the space-science preference. For Role B, these two policies selected La La Land most often, while epsilon-greedy also selected Toy Story in 43.73% of rounds, consistent with the profile accepting cartoons.

Part 7A. Role B selections and variability

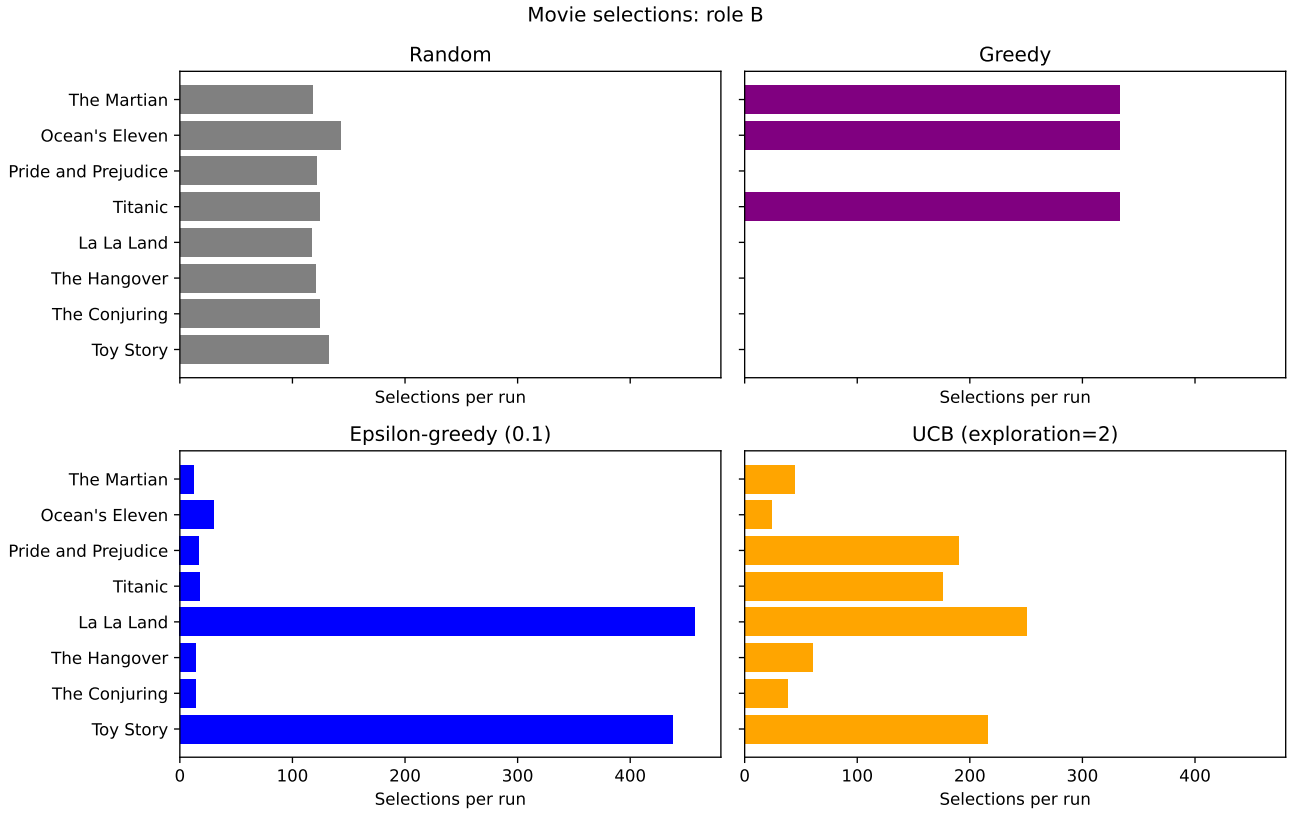


Figure 14. Part 7A: mean movie-selection counts per run for Role B.

Policy	Role A: most selected movie and mean share	Role B: most selected movie and mean share
Random	Ocean's Eleven (14.30%)	Ocean's Eleven (14.30%)
Greedy	Three-way tie (33.33% each)	Three-way tie (33.33% each)
Epsilon-greedy (0.1)	The Martian (82.30%)	La La Land (45.77%)
UCB	The Martian (60.33%)	La La Land (25.03%)

Greedy's three-way tie is across runs: it selected Titanic, Ocean's Eleven, or The Martian for all 1000 rounds in one run each, under both roles. This dependence on the initial choice explains its high variability and its lower mean than Random for Role B. Random's most-selected movie reflects sampling, not learned preference. No independent oracle established the optimal movie. With only three trajectories, the 10th-90th percentile bands describe observed variation and are not confidence intervals; these rankings apply to the specified personas, model, and seeds.

UCB uses the bonus sqrt(2 log(t + 1) / n) and tries every movie once. Exact ties are broken randomly. Identical choices share a fresh LLM score within that round; later rounds always obtain new scores.

Part 7B. Contextual LLM patient preferences

Five fixed ChatDoctor questions provide 17 distinct candidate responses from UltraMedical-Preference. Qwen2.5-0.5B last-token embeddings undergo a seed-42 random projection to 32 dimensions, normalization, centering/PCA to 16 components, common-radius scaling, and intercept addition, yielding 17 features. Qwen3.5-0.8B supplies score/10 rewards without added Gaussian noise (temperature 0.3). Profile A favors plain main advice and a clear next step; Profile B favors relevant detailed reasoning.

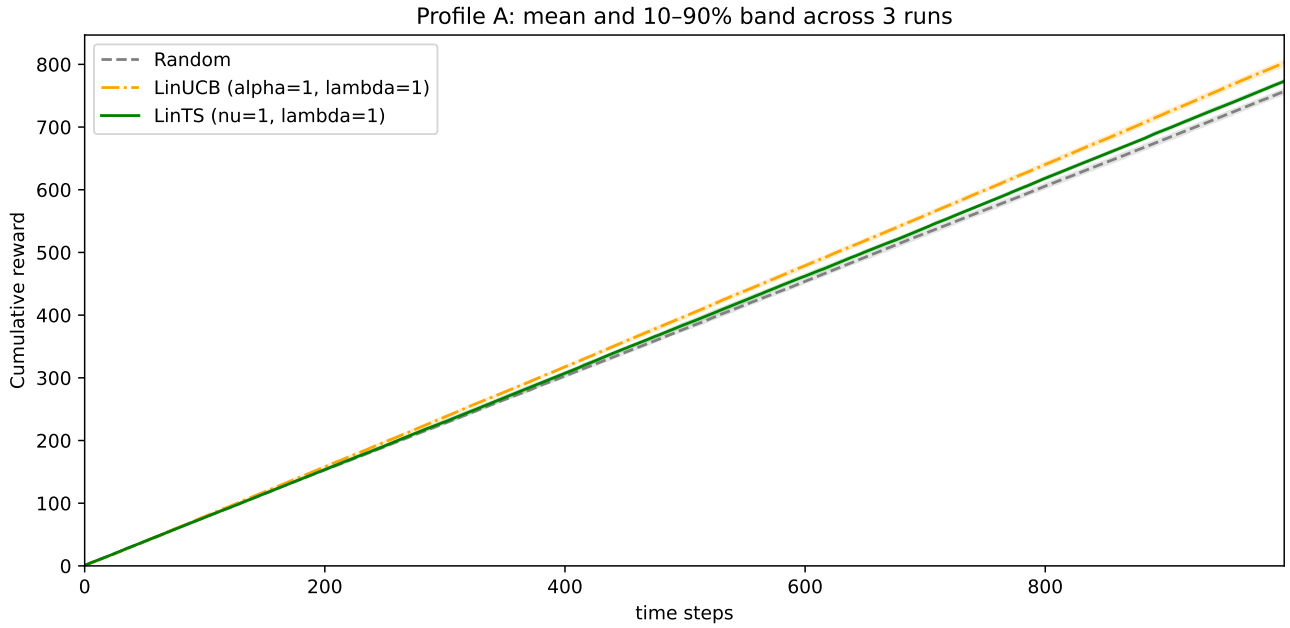


Figure 15. Part 7B: cumulative satisfaction reward for Profile A.

Final results: N = 3 and T = 1000 per profile

Policy	Profile A cumulative reward	A mean reward	Profile B cumulative reward	B mean reward
Random	756.87 ± 5.27	0.7569	889.73 ± 5.98	0.8897
LinUCB	802.57 ± 5.78	0.8026	945.73 ± 3.91	0.9457
LinTS	773.03 ± 2.62	0.7730	916.67 ± 2.24	0.9167

LinUCB had the highest observed mean reward for both profiles, with LinTS second. Its gains over Random were **45.70** cumulative reward for Profile A and **56.00** for Profile B, equivalent to 0.0457 and 0.0560 per round. These results favor LinUCB for this evaluated setting without establishing a universal algorithm ranking.

Part 7B. Profile B and interpretation

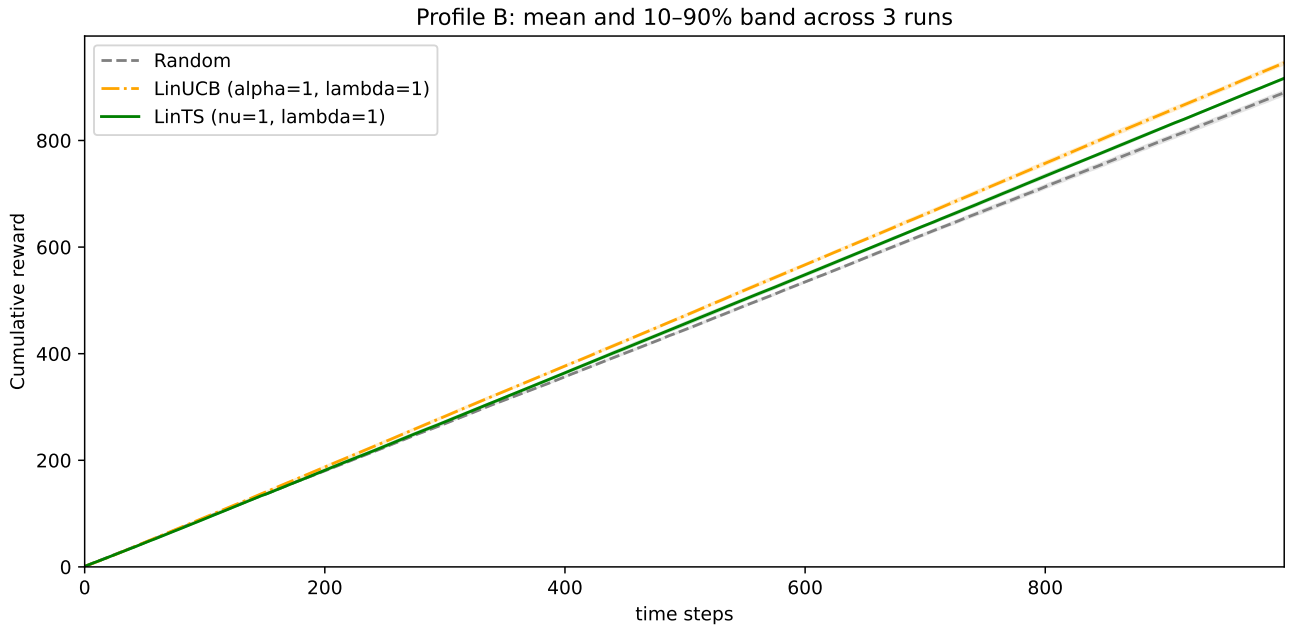


Figure 16. Part 7B: cumulative satisfaction reward for Profile B.

Model and comparison design

LinUCB uses alpha = lambda = 1. LinTS samples a shared linear coefficient with nu = lambda = 1. Each learning algorithm maintains its own ridge estimate across questions. All policies receive the same question in a given round; identical response choices share a freshly sampled score. Seed 7002 gives matching A/B question schedules. The Gaussian linear model approximates the bounded satisfaction reward.

Results and discussion

All policies received higher rewards under Profile B. This is consistent with the available responses matching its preference for detailed explanations, but changing the profile also changes the satisfaction judgment. The comparison therefore does not isolate a causal effect of answer length or establish clinical quality.

The study uses five fixed questions, 17 distinct responses, and fixed encoded/PCA features. Algorithms share questions and fresh same-round scores for identical choices; the A/B question schedules also match. With only three trajectories, the 10th-90th percentile bands describe observed variability rather than confidence intervals. Conclusions apply to these candidate responses, personas, model revisions, and seeds.

Reproducibility and modeling assumptions

Algorithm conventions

Linear TS uses a zero-mean Gaussian prior with precision 1 and unit observation-noise variance, drawing N(theta-hat, V-inverse). The supplied LinUCB radius is retained as a heuristic. LMCTS uses the noisy regularized-loss update below, with z standard normal.

$$\theta \leftarrow \theta - \eta(\text{grad MSE}(\theta) + \lambda\theta) + \sigma \sqrt{2\eta\beta^{-1}}z$$

For NeuralTS/UCB, normalized gradient features are phi = gradient(f)/sqrt(50), and the diagonal precision accumulates squared features. Training uses mean MSE plus lambda times 50 times the squared distance from initial weights divided by the observation count. These neural uncertainty and Langevin conventions are explicit implementation assumptions, not exact posterior guarantees.

Seeds and saved execution

Part	Environment / trajectory seeds
1-2	42 / 52; epsilon sweep 62; fresh comparison 63
3-4	72 / 73; Part 4 coefficient 82, geometry 83, trajectories 84
5-6	Part 5: 505; shuttle tuning 606; fresh comparison 607
7A-7B	Movie roles: 7001; patient profiles: 7002

Experiment source: **13f4e56dea345669e56e07da1dc06c2d2ea3f5ac**. Run archive: artifacts/full-20261101-1923. All 43 executed cells, original N/T, finite arrays, saved statistics, and 16 PNG/PDF pairs passed independent validation. Actual reward generation completed 18,479 movie and 12,757 patient calls with zero failures.

Pinned sources and software

Source	Pinned revision
Qwen/Qwen3.5-0.8B (both scorers)	2fc06364715b967f1860aea9cf38778875588b17
Qwen/Qwen2.5-0.5B (encoder)	060db6499f32faf8b98477b0a26969ef7d8b9987
TsinghuaC3I/UltraMedical-Preference	761eb7935310ba662a96d93c5af342e5269d5759

RL environment: Python 3.11.18; PyTorch 2.10.0+cu128; NumPy 2.4.6; pandas 3.0.6; scikit-learn 1.9.1; transformers 5.3.0; datasets 4.8.5. Shuttle was loaded through OpenML's active dataset version. Both LLM scorers used top-p 0.9 and top-k 20.

The companion code submission is Experiments.ipynb. Source and archived results: https://github.com/zhouhanqian1-sys/bandit-algorithms. The report uses existing notebook analyses and original vector figures; no raw medical response texts or model cache files are included.